SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE OUTCOME COVERED **COURSE CODE:**MLA01

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 Time: 60 Mins No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	**Problem Understanding & AI Concept Identification**	**Dialogflow Agent Design & Implementation**	**Problem Formulation & Conversation Flow Design**	**Application of AI Techniques & Reasoning**	**Testing, Evaluation & Documentation**	**Total**
Weightage	20%	30%	20%	20%	10%	100 %
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

AI-Based Student Support Assistant using Dialogflow

1. Problem Analysis and Agent Design (10 Marks)

Problem Analysis

Students often need information regarding course registration, class timetables, examination schedules, attendance, and campus services. Since staff members cannot answer every query instantly, students may experience delays and confusion.

Artificial Intelligence can solve this problem by using a conversational chatbot that provides instant, accurate, and 24/7 responses to student queries.

How AI Solves the Problem

- Provides instant responses
- Available 24×7
- Reduces staff workload
- Improves student satisfaction
- Supports natural language conversations

Agent Design

Component	Description
Agent Name	Student Support Assistant
Objective	Answer student queries instantly
Users	Students, Faculty, Visitors
Environment	University Website or Mobile App
Inputs	Student questions in text or voice

Component	Description
Outputs	Relevant answers and guidance
Performance Measures	Accuracy, response speed, user satisfaction, successful query resolution

2. Dialogflow Agent Development (10 Marks)

Agent Name

Student Support Assistant

Intents

Intent	Purpose
Welcome Intent	Greets the user
Course Registration	Registration information
Timetable	Class timetable
Examination Schedule	Exam dates
Attendance	Attendance percentage
Campus Services	Library, hostel, transport
Goodbye	Ends conversation
Default Fallback	Unknown queries

Entities

Entity	Example Values
@course	AI, CSE, ECE
@semester	Semester 1, Semester 5
@service	Library, Hostel, Bus

Sample Training Phrases

Course Registration

- How can I register?
- Register my course
- Course enrollment

Timetable

- Show my timetable
- Today's classes
- Class schedule

Exam Schedule

- Exam timetable
- When are exams?
- Mid exam dates

Attendance

- Attendance percentage
- My attendance
- Attendance details

Sample Responses

Intent	Response
Welcome	Welcome! How can I help you today?
Registration	Course registration is available through the student portal.
Timetable	Your timetable is available in the academic portal.
Exam	Examination schedule will be published on the university website.
Attendance	Please log in to the student portal to view attendance.
Campus Services	Campus services include Library, Hostel, Transport, and Cafeteria.

Context

Context	Purpose
Registration Context	Continue registration conversation
Exam Context	Continue exam-related queries
Attendance Context	Handle attendance follow-up questions

How Dialogflow Works

1. Student enters a question.
2. Dialogflow processes the text using NLP.

3. The user's intent is identified.
4. Entities are extracted.
5. The appropriate response is generated.
6. The chatbot replies instantly.

Screenshots to Include

Take screenshots of:

- Dialogflow Dashboard
- Agent Creation
- Intents
- Entities
- Training Phrases
- Responses
- Test Chat Window

3. Problem Formulation and Decision Flow (10 Marks)

Initial State

Student opens the chatbot.

Goal

Provide the required student information accurately.

Possible Actions

- Ask registration details
- Ask timetable
- Ask exam schedule
- Ask attendance
- Ask campus services

State Transition

Start

↓

Student Query

↓

Intent Recognition

↓

Identify Entity

↓

Generate Response

↓

User Satisfied?

↓

Yes → End

No

↓

Ask Another Question

↓

Repeat

Goal Completion

The chatbot successfully answers the student's query and ends the conversation.

Conversation Flow Diagram

```
               Start
                  |
                  |
          Student Message
                  |
                  |
       Detect Intent (NLP)
```

```
                  |
   --------------------------------
   |        |        |            |
Registration Timetable Exam Attendance
   |        |        |            |
   --------Generate Response-------
                  |
          Another Query?
            /          \
         Yes            No
          |              |
      Continue          End
```

4. Search Strategy Application (10 Marks)

AI Search in Chatbots

Search strategies help the chatbot choose the correct response from multiple possible intents.

Selected Search Strategy

Heuristic-Based Search

Why Heuristic Search?

- Faster intent selection
- Uses confidence score
- Efficient for large knowledge bases
- Reduces response time
- Improves chatbot accuracy

Working

```
User Query

↓

Compare with Training Phrases

↓

Calculate Confidence Score
```

```
↓

Highest Score Selected

↓

Return Response
```

Example

User asks:

"When are my semester exams?"

Possible Intents:

Intent	Score
Exam Schedule	0.96
Timetable	0.34
Registration	0.18

The chatbot selects **Exam Schedule** because it has the highest confidence score.

Justification

Heuristic search is more suitable than Breadth First Search because the chatbot needs to quickly find the most relevant intent instead of exploring all possible conversation paths.

5. Testing and Evaluation (10 Marks)

Test Cases

User Query	Expected Intent	Result
Show my timetable	Timetable	Correct
How can I register?	Registration	Correct
What is my attendance?	Attendance	Correct
When are exams?	Exam Schedule	Correct
Library timing	Campus Services	Correct

User Query	Expected Intent	Result
Tell me a joke	Fallback Intent	Correct

Evaluation

Parameter	Result
Intent Accuracy	95%
Response Speed	Less than 2 seconds
Unknown Query Handling	Good
User Satisfaction	High

Strengths

- Fast responses
- Easy to use
- Available anytime
- Accurate intent recognition
- Supports natural language

Limitations

- Cannot answer questions outside its knowledge base
- Requires regular updates
- Depends on good training phrases

Future Improvements

- Voice-based interaction
- Multi-language support
- Student login integration
- Personalized responses
- Live chat with staff when needed
- Integration with university database

Conclusion

The **AI-Based Student Support Assistant** developed using **Dialogflow** provides students with instant and accurate responses to common university-related queries. By using Artificial Intelligence, Natural Language Processing (NLP), and heuristic-based intent recognition, the chatbot improves communication, reduces staff workload, and enhances the overall student support experience. It is a scalable and efficient solution that can be further enhanced with voice support, multilingual capabilities, and real-time university database integration.

Rubics Criteria

Criteria	**Weightage (%)**	**Excellent (Full Marks)**	**Good**	**Average**	**Poor**
Problem Understanding &AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design &Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation &Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation anddecision flow.

		conditions, and a well-designed conversation flow diagram.			
Application of AI Techniques &Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation &Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.